The Fourier Series for a Square Wave and the Sinc Function



Copyright © Barry Van Veen 2015

AllSignalProcessing.com



Key Concepts

1) Square wave with fundamental period T

$$x(t) = \begin{cases} 1, & |t| < T_o \\ 0, & T_o < |t| < T/2 \end{cases}$$

2) Fourier series coefficients

$$x(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} X[k] = \frac{2\sin(k\Omega_o T_o)}{k\Omega_o T} = \frac{\sin(k2\pi T_o/T)}{k\pi}$$

3) As T_o/T decreases,

a) Each period of square wave has narrower non zero portion.

b) The Fourier series coefficients get wider in the frequency index k.

c) The amplitude of the Fourier series coefficients decrease.

4) The sinc function $\text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}$ occurs frequently in Fourier analysis.

a) The value at $u = 0$ is defined as the limit as $u \to 0$ and is equal to 1.

b) The main lobe is between the first zero crossings, that is $-1 < u < 1$.

c) Sidelobe are due to oscillations of the sine in the numerator for $|u| > 1$.

d) Sidelobe peak amplitudes decay inversely proportional to u.

5) Partial-sum approximation to the square wave. Define

$$x_J(t) = 1/2 + \sum_{k=1}^{J} B[k] \cos(k\Omega_o t)$$

a) $x_J(t) = 1/2$ at the discontinuities of the square wave

b) $x_J(t)$ exhibits ripples called Gibbs' phenomenon on both sides of discontinuities. The ripples are about 9% the size of the discontinuity no matter how big J becomes.

AllSignalProcessing.com

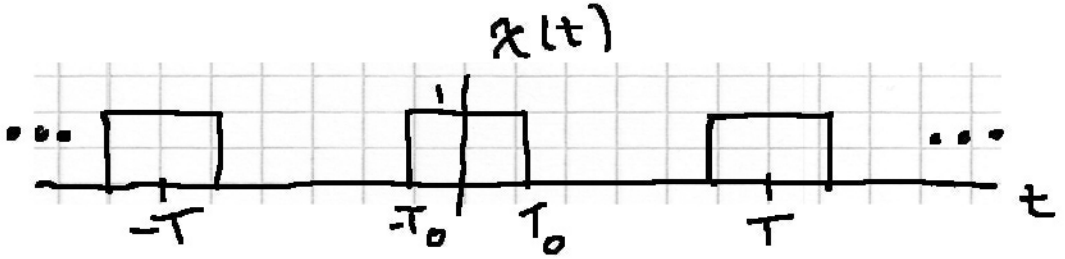
Video Screenshots

Fourier Series: Periodic, continuous-time signals

$$x(t) = \sum_{k=-\infty}^{\infty} X[k] e^{jk\Omega_0 t} \quad \overset{FS;\,\Omega_0}{\longleftrightarrow} \quad X[k] = \frac{1}{T}\int_{\langle T\rangle} x(t) e^{-jk\Omega_0 t}\,dt$$

$$\Omega_0 = \frac{2\pi}{T}$$

Square Wave:



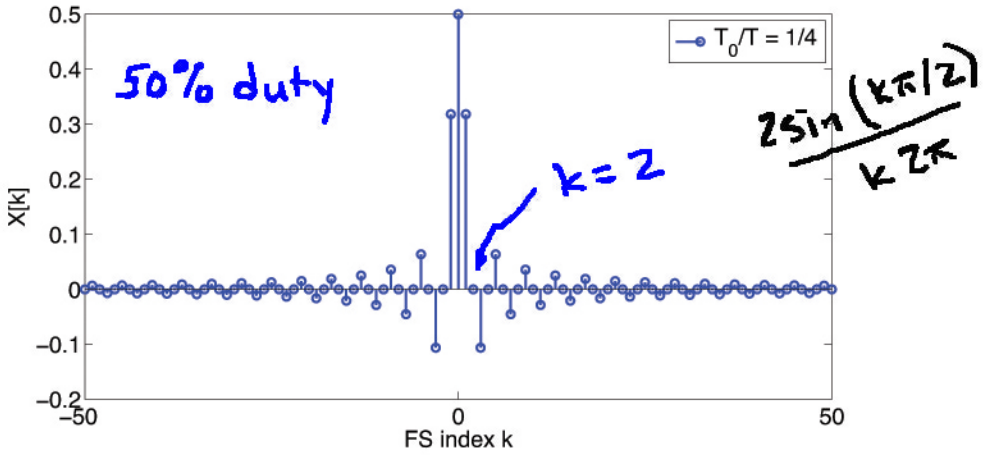
$$X[k] = \frac{1}{T}\int_{-T_0}^{T_0} 1\cdot e^{-jk\Omega_0 t}\,dt = \frac{-1}{jk\Omega_0 T} e^{-jk\Omega_0 t}\Big|_{-T_0}^{T_0}, \; k\neq 0 \qquad X[0] = 2\frac{T_0}{T}$$

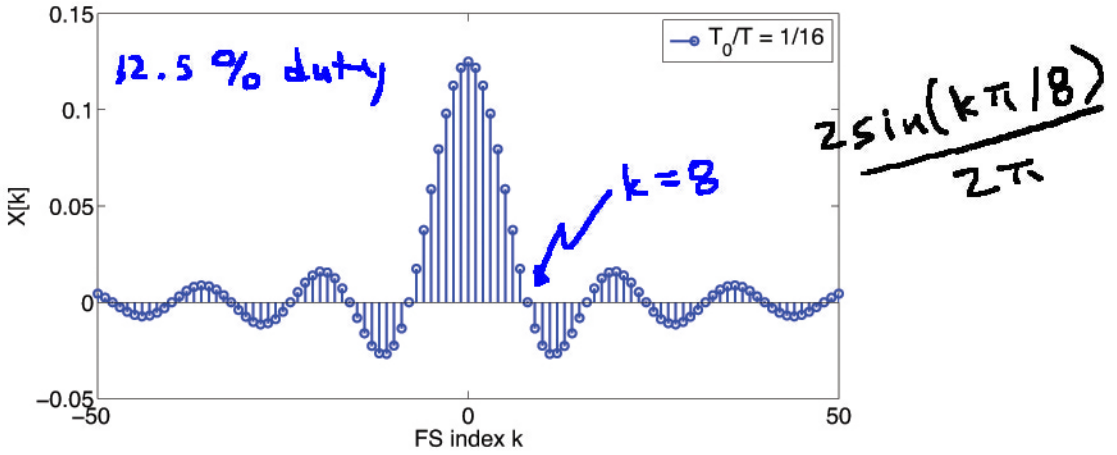
$$= \frac{2}{k\Omega_0 T}\left(\frac{e^{jk\Omega_0 T_0} - e^{-jk\Omega_0 T_0}}{2j}\right) = \frac{2\sin(k\Omega_0 T_0)}{k\Omega_0 T}, \; k\neq 0$$

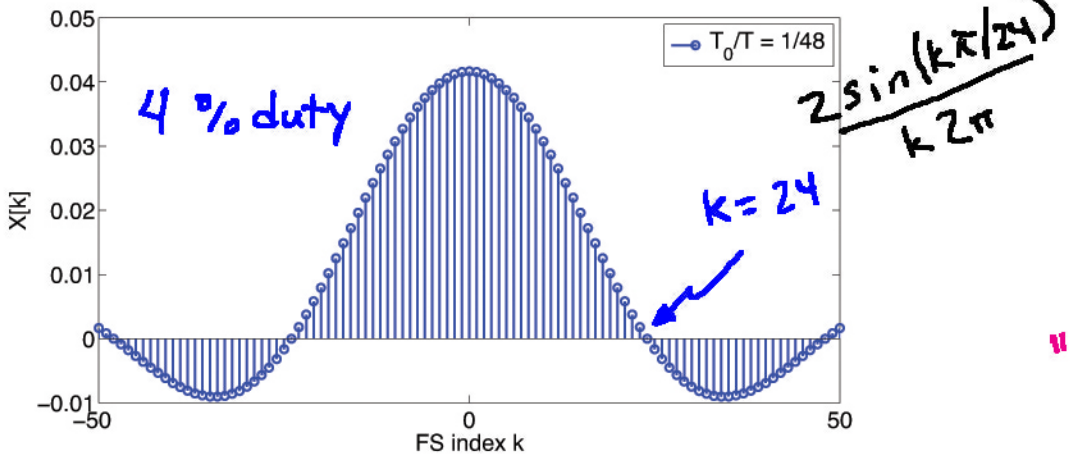
Note: $\lim_{k\to 0} \frac{2\sin(k\Omega_0 T_0)}{k\Omega_0 T} = \frac{2T_0}{T}$

$$x(t) \overset{FS;\,\Omega_0}{\longleftrightarrow} X[k] = \frac{2\sin(k\Omega_0 T_0)}{k\Omega_0 T}$$

$$X[k] = \frac{2\sin(k\Omega_0 T_0)}{k\Omega_0 T} = \frac{2\sin\left(k2\pi\left(T_0/T\right)\right)}{k2\pi}$$





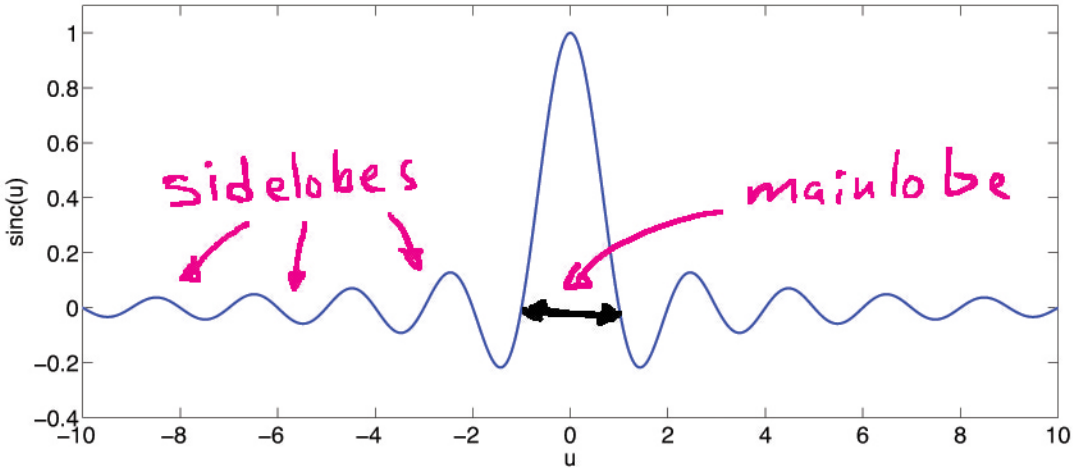


As T_0/T decreases
- narrower in time
- amplitude decreases
- wider in frequency

"width" in time and frequency are inversely related

The sinc function

$$\text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}$$



1) Max value $\lim_{u \to 0} \frac{\sin(\pi u)}{\pi u} = 1$

2) Oscillates due to $\sin(\pi u)$

3) zero crossings at integer u

4) Decays like $\frac{1}{\pi u}$

(Be alert to other definitions)

Square wave

$$X[k] = \frac{2\sin(k\Omega_0 T_0)}{k\Omega_0 T}$$

$$= \frac{2\sin\left(\pi k \frac{2T_0}{T}\right)}{\pi k 2}$$

$$= \frac{2T_0}{T} \frac{\sin\left(\pi k \frac{2T_0}{T}\right)}{\pi \left(k\frac{2T_0}{T}\right)} = \frac{2T_0}{T}\text{sinc}\left(k\frac{2T_0}{T}\right)$$

u

Square Wave

$$X[k] = \frac{2\sin(k\Omega_0 T_0)}{k\Omega_0 T}$$

real $\Rightarrow B[k] = 4\frac{\sin(k\Omega_0 T_0)}{k\Omega_0 T}$

$A[k] = 0$

$$x(t) = \frac{2T_0}{T} + \sum_{k=1}^{\infty} \frac{4\sin(k\Omega_0 T_0)}{k\Omega_0 T}\cos(k\Omega_0 t)$$

Special Case: $T_0/T = 0.25$, 50% duty cycle

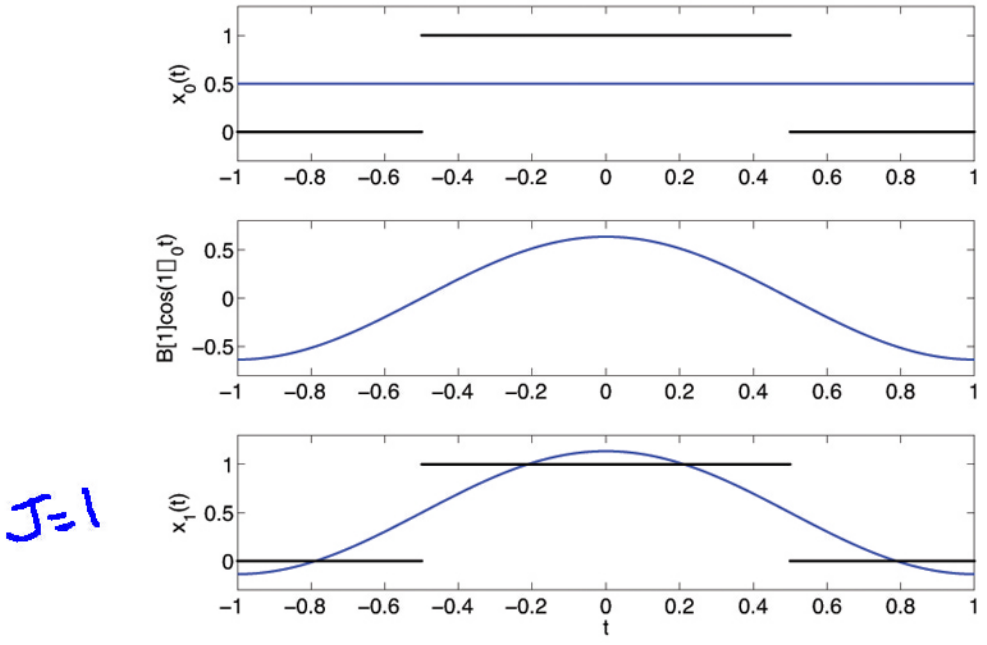
$$\sin(k\Omega_0 T_0) = \sin\left(k\frac{\pi}{2}\right) = \begin{cases} (-1)^{(k-1)/2} & k \text{ odd} \\ 0 & k \text{ even} \end{cases} \Rightarrow B[k] = \begin{cases} \frac{2(-1)^{(k-1)/2}}{\pi k} & k \text{ odd} \\ 0 & k \text{ even} \end{cases}$$

$$x(t) = \frac{1}{2} + \sum_{k=1}^{\infty} B[k]\cos(k\Omega_0 t) \rightarrow \text{odd harmonics}$$

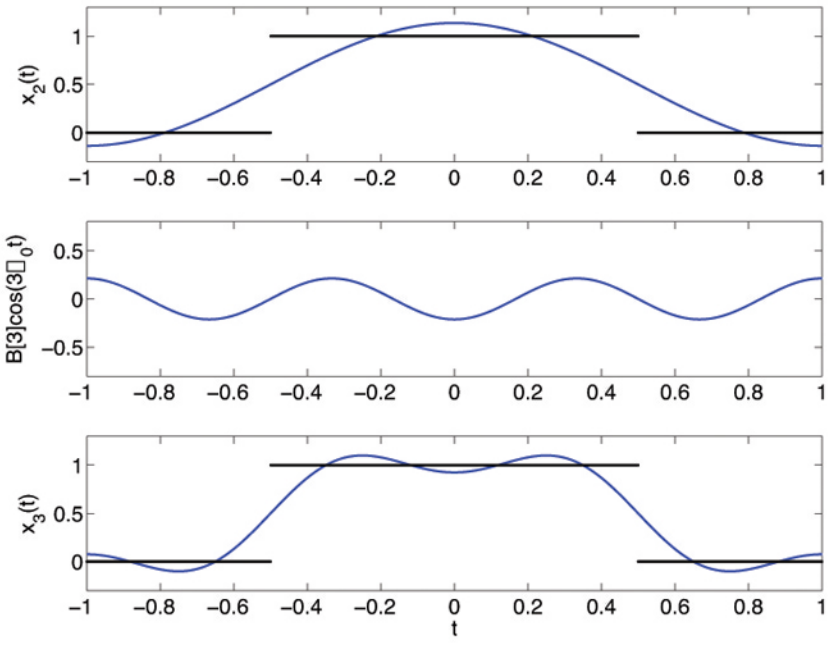
Partial Sum Approximation

$$x_J(t) = \frac{1}{2} + \sum_{k=1}^{J} B[k]\cos(k\Omega_0 t)$$

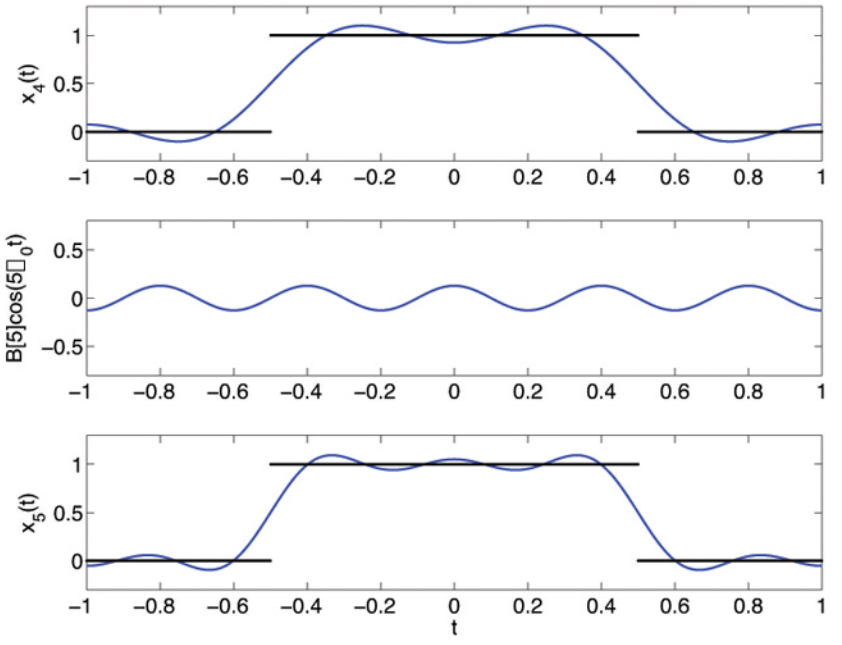
how does $x_J(t) \rightarrow x(t)$?


x0(t)
B[1]cos(1□0t)
x1(t)
t

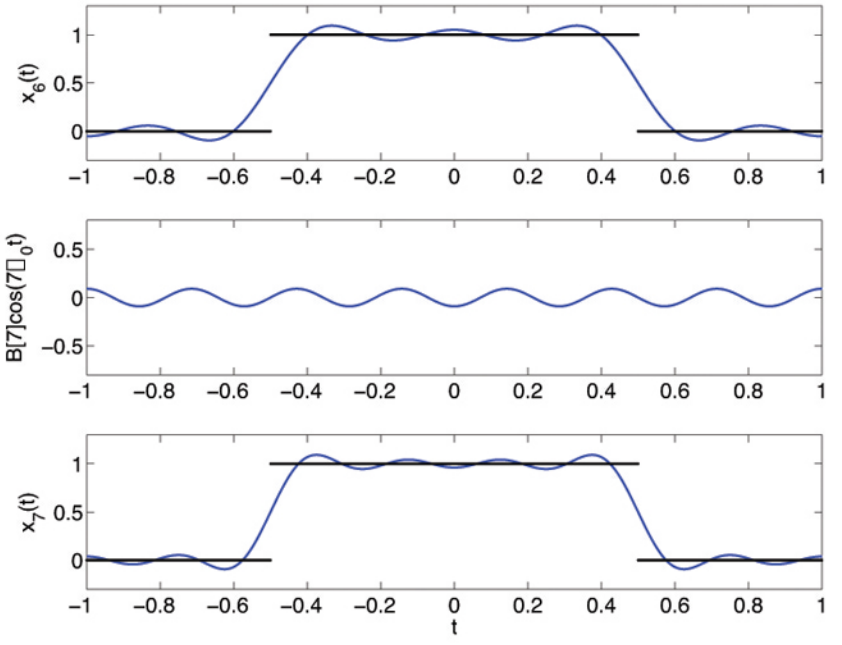
J=1


x2(t)
B[3]cos(3□0t)
x3(t)
t

J=3

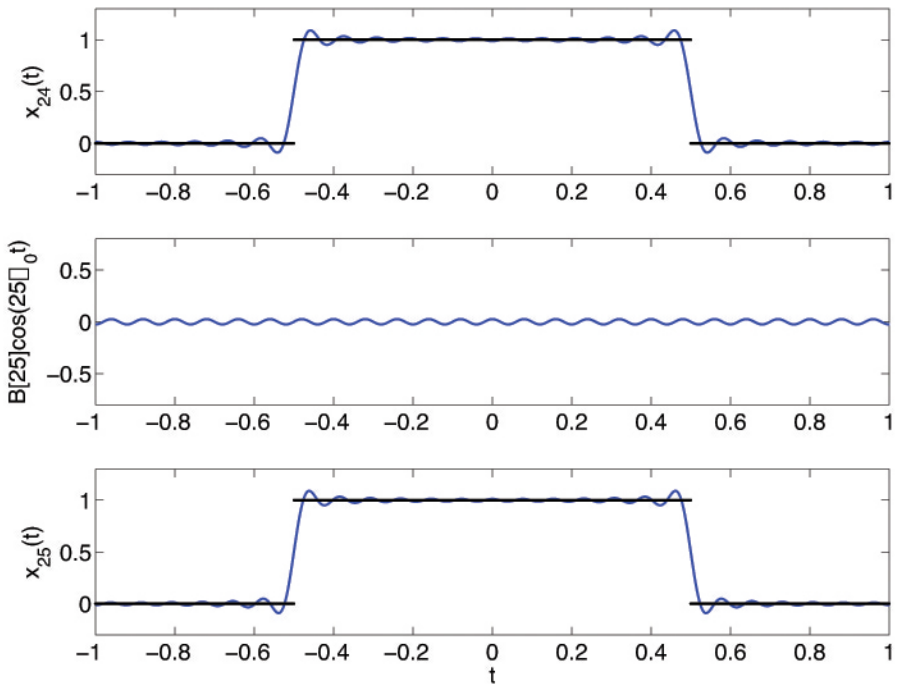

x4(t)
B[5]cos(5□0t)
x5(t)
t

J=5

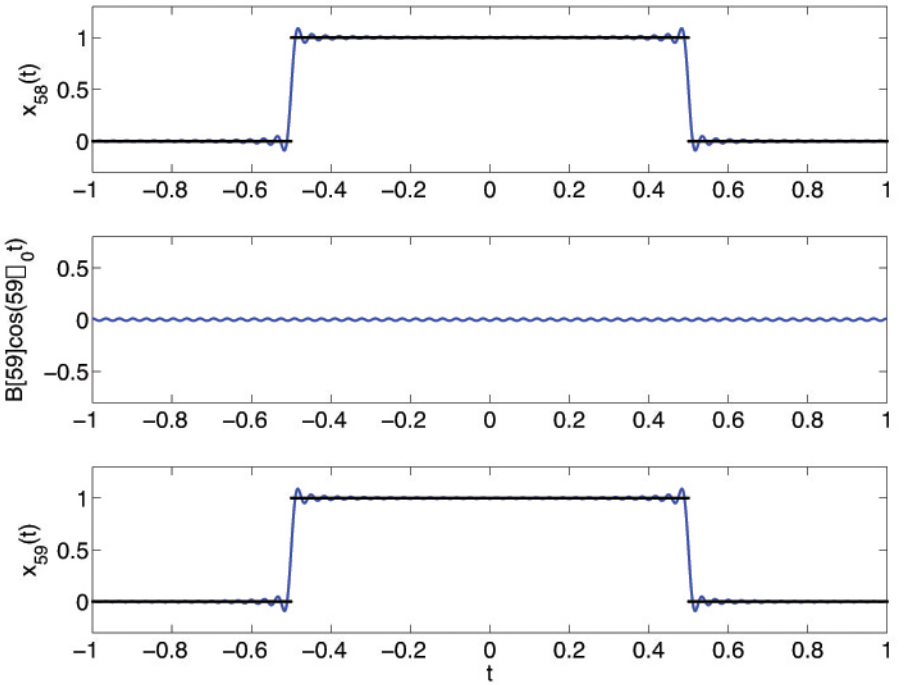

x6(t)
B[7]cos(7□0t)
x7(t)
t

J=7

5



J=25



J=59

- $x_J(t) = 0.5$ for $t = \pm 1/2$
- Ripple near discontinuity "Gibbs' phenomenon"
  - about 9% peak no matter how big J becomes
  - due to truncation of sum

Copyright 2015
Barry Van Veen

